FINTECHATHON 2026 · TOPIC E · PROJECT PRESENTATION

Setu (সেতু)

The bridge between a Bangladeshi importer and a Chinese supplier: financing, cash flow and safe payment, in one AI copilot.

The AI explains. Code decides.

Simulated sandbox

No model trained

PROBLEM

Three ways a China import goes wrong

①

Cash gap, unseen

Goods, freight, duty, VAT, bank charges and FX add up. The shortfall is found when a milestone is already due, and production stops.

②

Document fraud

A forged bill of lading, or one invoice financed at two lenders, can cost the whole advance. Hidden PDF text can even command an AI.

③

Redirected payment

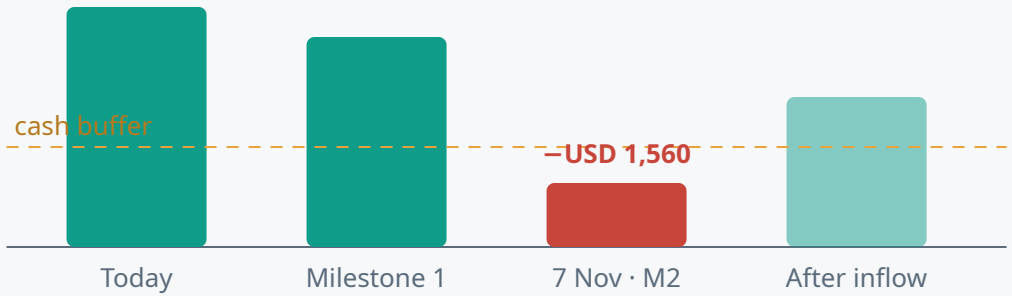
A "new bank account" arrives by email. Licence, invoice and account names no longer match, and the money lands with a stranger.

SMEs need an assistant that can reason about the whole order, and can never be talked into moving money.

SCENARIO

Rahima's order: USD 11,200 landed, USD 6,000 in hand

Cash-flow forecast for order SETU-2026-0001



Illustrative shape of the seeded demo; the shortfall figure is computed by the cash-flow engine.

USD 1,560

SHORTFALL alert on 7 Nov 2026 (milestone 2), raised by deterministic code, T-7 / T-3 / T-1

USD 5,200

order financing gap: 11,200 landed cost − 6,000 available cash

1 of 4

suppliers is suspicious: licence name ≠ receiving-account name, personal, new account

SOLUTION

Nine Topic E requirements, three pillars

Plan the money

- **FR-1** Financing match, ≥3 options
- **FR-2** 90-day cash flow and alerts
- **FR-3** Landed cost, margin, scorecard
- **FR-7** USD vs CNY route, rate lock

Trust the counterparty

- **FR-4** Supplier KYB name matching
- **FR-5** Milestone escrow, 30/60/10
- **FR-6** Document forensics, hash registry
- **FR-8** Signed credit record

Ask in your language

- **FR-9** Chat in English and Bangla (Chinese for suppliers)
- Every answer cites the numbers and rules used
- Cards show cash flow, financing, tiers

Where something is mock, we say so: lenders are fictional, FX spreads are mock, and all money movement is labelled **Simulated**.

LIVE DEMO

One order, from question to signed record

1

Ask in Bangla

Streamed answer with a cash-flow card: USD 1,560 short on 7 Nov.

2

Compare financing

Supplier credit chosen. The escrow-backed Lender B loan is blocked: the invoice is already pledged at Lender A.

3

Tier 2 one-tap release

Tranche 1 paid in CNY with the FX saving, simulated.

WOW A

Forged bill of lading

Check FAILS, hidden white text revealed, rule trace shows Tier 3. Chat says it ignored the instruction.

WOW B

Duplicate invoice

The same invoice offered at Lender B is blocked by the shared hash registry.

6

Signed credit record

Escrow completes; an Ed25519 record is issued and the audit chain verifies.

Everything shown runs through the real API. A recorded-reply fallback and a pre-flight checklist cover network or model outages.

SECURITY DESIGN

Risk tiers decided by code, approved by humans

Tier 0 · Auto
Analysis and checks

Tier 1 · Auto + notice
Verified, unchanged beneficiary, checks pass, ≤ USD 20,000

Tier 2 · One-tap approval
First deal, minor mismatch, FX lock, schedule change

Tier 3 · OTP + 10-min cooling-off
Beneficiary change, high fraud score, duplicate hash, > USD 20,000, instruction found in a document

LLM can only propose

The model is offered Tier 0/1 tools only, and a test asserts it.

Document text is data

Instruction-like text, even hidden, escalates to Tier 3.

Privacy Gateway

NID, accounts, phones and prices are masked before any model call.

Signed audit chain

Every action is logged, hash-chained and verifiable.

EVIDENCE

Measured results

Development split, synthetic data. Every number is reproducible with the evaluation harness.

1.00

forensics precision and recall; 11 checks, 10 fraud types; 0/50 clean packs flagged

5/5

duplicate-financing attempts caught: 3 exact, 2 re-saved

100→0%

injection attack success, defences off → on (13 attacks); 0/300 false positives

0

detectable identifiers sent to the model with masking on (2,490 with it off)

Why is dev perfect? The data is synthetic and the generator and checks share a team, so these are ceilings, not real-world accuracy. The injection harness uses an obedient mock model. The test split is run once and reported as is.

INNOVATION

Trust you can see and verify

Explained risk tiers

Every decision shows its rule trace and "what would change it" hints.

Privacy X-ray

See exactly what the model received, masked tokens included.

Live attack console

Upload a hidden-instruction document and compare defences off vs on.

Public verification

Anyone can check the audit log or a credit record with a public key, in the browser, with no login.

ARCHITECTURE

A thin AI layer over deterministic finance code

Next.js web app
talks only to the API · EN / BN / ZH

▼ REST + SSE

FastAPI backend
approvals, escrow, auth, audit endpoints

▼

PostgreSQL + pgvector (Neon in production)

LangGraph agent
router → Tier 0/1 tools → explainer

LLM client (only model entry point)
Privacy Gateway → cache → Claude Haiku / Sonnet

Domain code (no LLM)
landed cost · financing · cash flow · FX · escrow · KYB · forensics

Security layer
tier engine · OTP approvals · injection detector · signed audit

LIMITS, ROADMAP, CLOSE

Honest about today, clear about next

Limits

- Synthetic data; dev-split numbers
- Fictional lenders, mock FX spreads
- Sandbox money only
- Bangla and Chinese not natively reviewed

Roadmap

- Report the held-out test split
- Real authentication and SMS for OTP
- PayPal sandbox escrow, lender integrations
- Native-speaker review; pilot with importers

Setu: fewer cash surprises, fewer fraud losses, and payments a human can always trust.

Thank you · Questions welcome